The effect of deviations from the Kerr metric on the fluorescent Fe Kα line as a test for general relativity.

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1 INTRODUCTION

The Johannsen metric Johannsen (2013) allows for deviations from the standard Kerr hypothesis describing a rotating black hole. The deviations are permitted by the deformation parameters α_{13}, ϵ_3, α_{22}, and α_{52}. α_{13} and ϵ_3 have both independently been used to conduct strong field tests of general relativity through fitting to the Fe Kα emission line from active galactic nuclei Bambi et al. (2017); Cao et al. (2018); Tripathi et al. (2019, 2026), however they have not been tested simultaneously, which was the goal of this project.

2 THE PARAMETER SPACE OF THE JOHANNSEN METRIC.

To define a fitting model from the Johannsen metric it was necessary to first understand the parameter space. The relativistic raytracing code `Gradus.jl` Baker & Young (2025b) was used to explore the metric by computing the observed emission line profile, which is significantly broadened by general relativistic effects Fabian et al. (2000). Gradus was also used to render images of an accretion disk around the black hole to observe the redshift. The key findings from this were:

(i) The radius of the ISCO increases for increasing α_{13} and decreases for increasing ϵ_3.

(ii) The orbital frequency for a test particle increases for increasing α_{13} leading to an increase in the degree of observed redshifting Johannsen (2013). The opposite effect occurs for ϵ_3.

(iii) For $\alpha_{13} \lesssim 5.7$ the behaviour of the ISCO becomes chaotic due to the presence of two minima in the potential. This led to unpredictable behaviour in this region.

(iv) For $\alpha_{13} \lesssim 6.2$ the energy and orbital frequency of an orbiting particle become imaginary leading to further unpredictable behaviour Johannsen (2013).

Johannsen defined a lower limit for both of these parameters as

$$\alpha_{13},\ \epsilon_3 > -\frac{(M+\sqrt{M^2-a^2})^3}{M^3}, \tag{1}$$

for a black hole with mass M and spin parameter a. This boundary ensures that the exterior region of the black hole has no singularities or time-like curves Johannsen (2013). It was found that there exists four additional unphysical regions of the Johannsen parameter space that exist only when α_{13} and ϵ_3 are varied simultaneously. Two of these regions yield naked singularities. The other two yield a metric with no valid ISCO (as calculated by `Gradus.jl`. This is likely due to the presence of two local minima in the potential of an orbiting particle.

These regions were not as easily parameterised, however three of them were found to take triangular shapes in the 2D plane created by the two deformation parameters. These regions could thus be bounded by linear equations with slopes and intercepts dependent on the spin parameter, a, and these equations used to define exclusion regions for the model.

3 FITTING TO OBSERVATIONS

A five-dimensional data table was constructed using `Gradus.jl` by varying the parameters a, h, θ, α_{13}, and ϵ_3, being the spin, corona height, inclination, and two deformation parameters, respectively. The grid size along each dimension was 10, ?, 9, 9, 10, respectively. For parameter combinations in the defined exclusion regions, an algorithm was defined to find the nearest "safe" combination in parameter space, and the line profile was calculated for this combination in place of the invalid one.

Data was used from the XRISM observatory XRISM Science Team (2020). [ENTER DETAILS HERE].

Fitting was completed using `SpectralFitting.jl` Baker & Young (2025a), the fit model was defined as

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Norm * Abs * (PowerLaw + Conv(XILLVER))
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where `Norm` is a normalisation constant, `Abs` models photoelectric absorption, `PowerLaw` describes the underlying power law continuum, and `XILLVER` models the reflection spectrum from the accretion disk García & Kallman (2010); García et al. (2013, 2014). The reflection spectrum was convolved by the table model defined above to give the relativistic spectrum.

- Fitting results

4 CONCLUSIONS

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DATA AVAILABILITY

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